

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)	(i)		Momentum = 40×2 (1) Momentum = 80 [kg m/s] (1)	1	1		2	2	
		(ii)		150 + 80 (ecf) = 230 [kg m/s]		1		1		
		(iii)		230 (ecf) [kg m/s]	1			1		
		(iv)		Velocity = $\frac{230 \text{ ecf}}{90}$ (1) Velocity = 2.56 or 2.6 [m/s] (1) Answer only of 2.5 or 2.55 [m/s] award 1 mark	1	1		2	2	
		(v)		$x = \frac{1}{2} (2.56 \text{ (ecf)} + 0) \times 25$ (1) $x = 32$ [m] (1) Answer only of 33 [m] award 2 marks	1	1		2	2	
	(b)			Abi's momentum {doubles / is 300 [kg m/s]} (1) Rachel's momentum {stays same / is 80 [kg m/s] / hasn't doubled} or Rachel's velocity hasn't doubled (1) [so incorrect] Accept total momentum = 380 [kg m/s so disagree] (2) Award 2 marks for to double the total momentum {Rachel's / both} velocities must double			2	2		
				Question 5 total	4	4	2	10	6	0